CUSTOMER CHURN & RETENTION ANALYSIS

Mini Project Report

Objective: Analyze customer demographics, services, contracts, billing and churn behavior and identify patterns associated with customer churn.

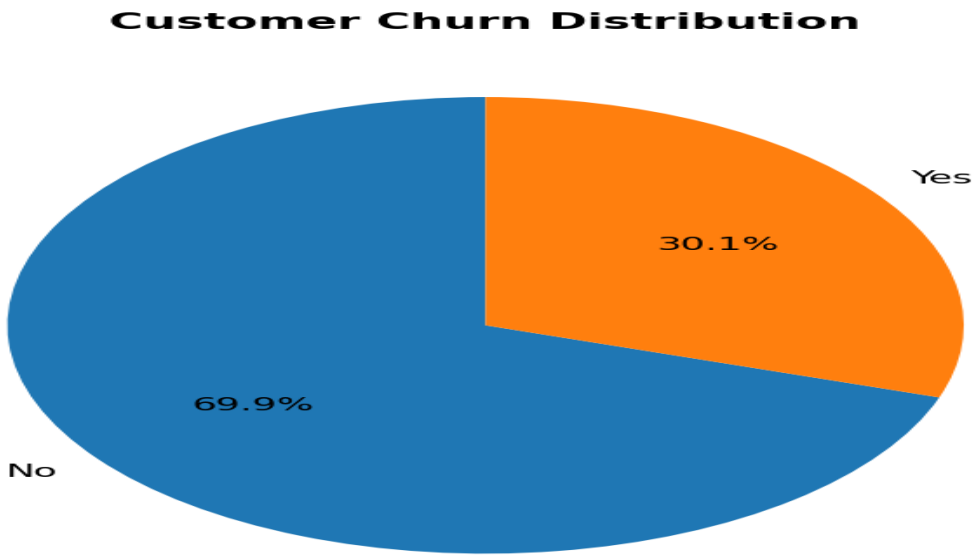
Dataset: Synthetic telecom/subscription customer dataset created for this academic mini-project with ChatGPT assistance. It follows the assignment suggested schema and contains 7,043 customer records. The assignment does not require IBM or another named dataset.

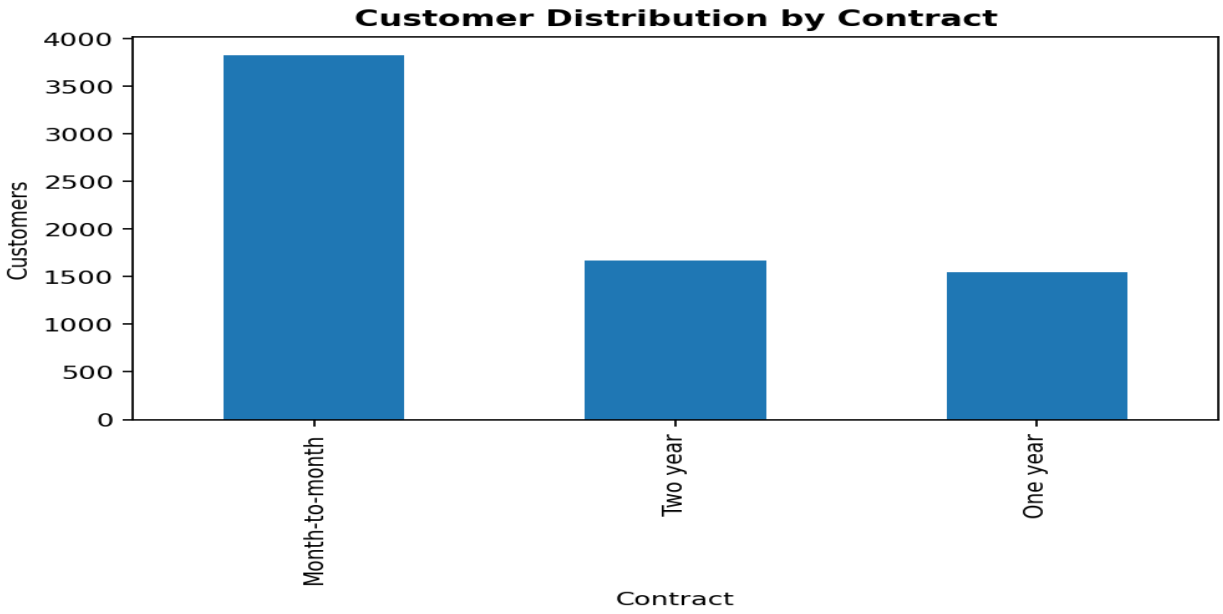
Metric	Result
Total Customers	7,043
Churned Customers	2,121
Retained Customers	4,922
Overall Churn Rate	30.12%
Average Monthly Charges	74.02
Average Total Charges	2367.50
Average Tenure	31.94 months

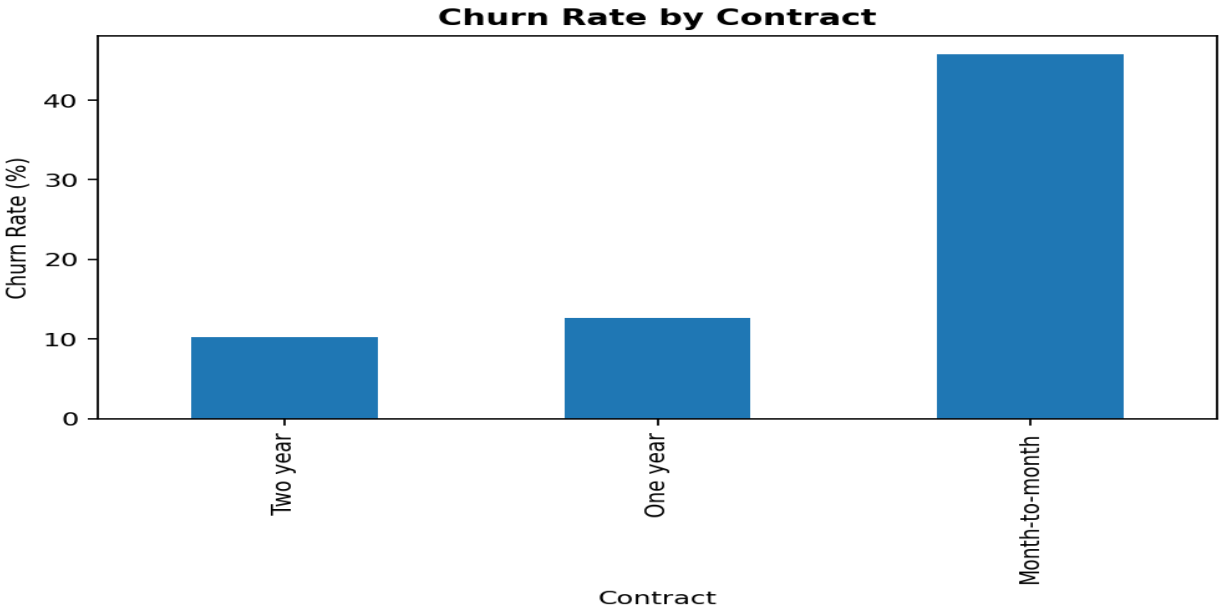
Required Analysis

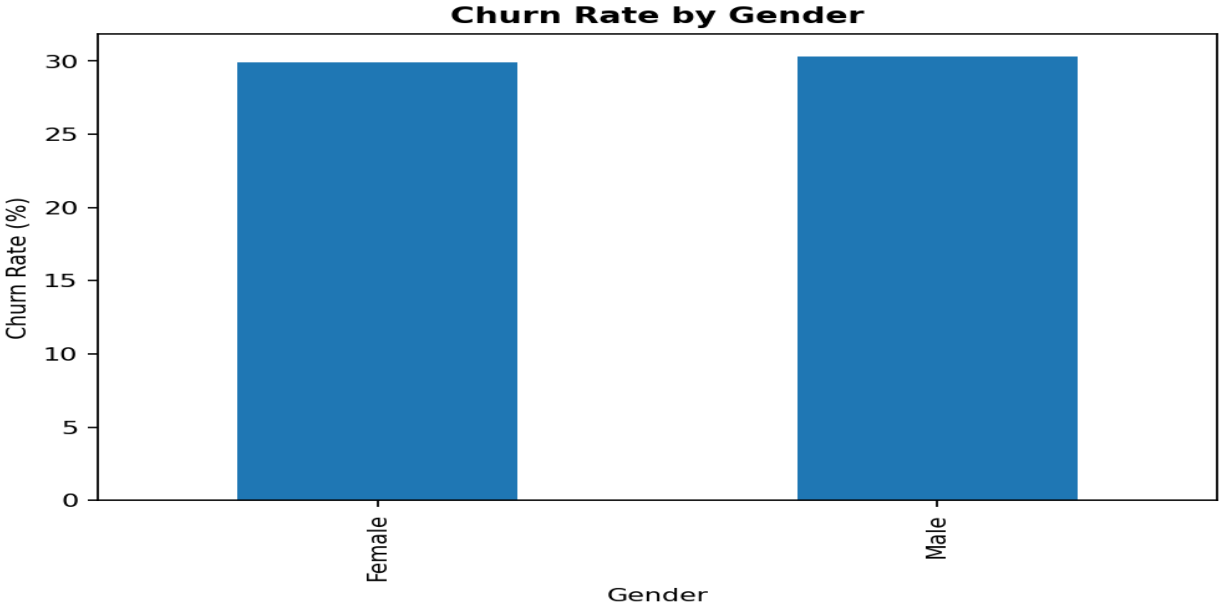
The Python notebook covers data loading, data types, descriptive statistics, missing values, duplicates, unique values, cleaning, tenure groups, all requested KPI calculations, segment distributions, churn rates by gender/contract/internet/payment/senior status/tenure group, churned-vs-retained charge and tenure comparisons, top 10 customers by total charges, high-churn segments and factors associated with churn.

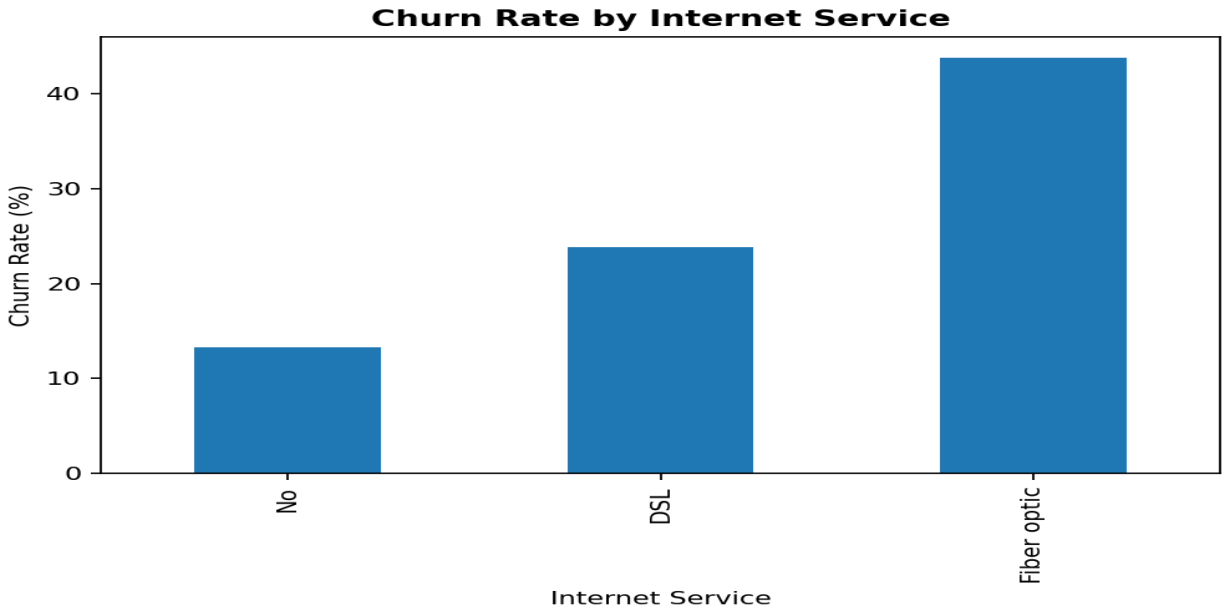
Python Visualizations — all 14 required charts are generated from the project dataset.

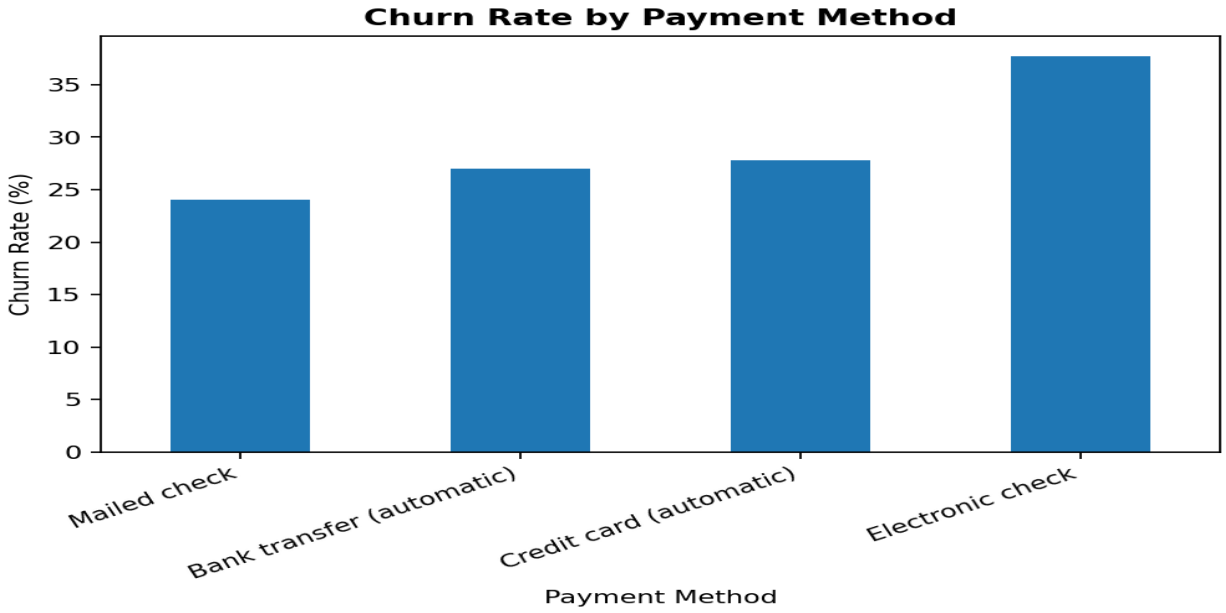


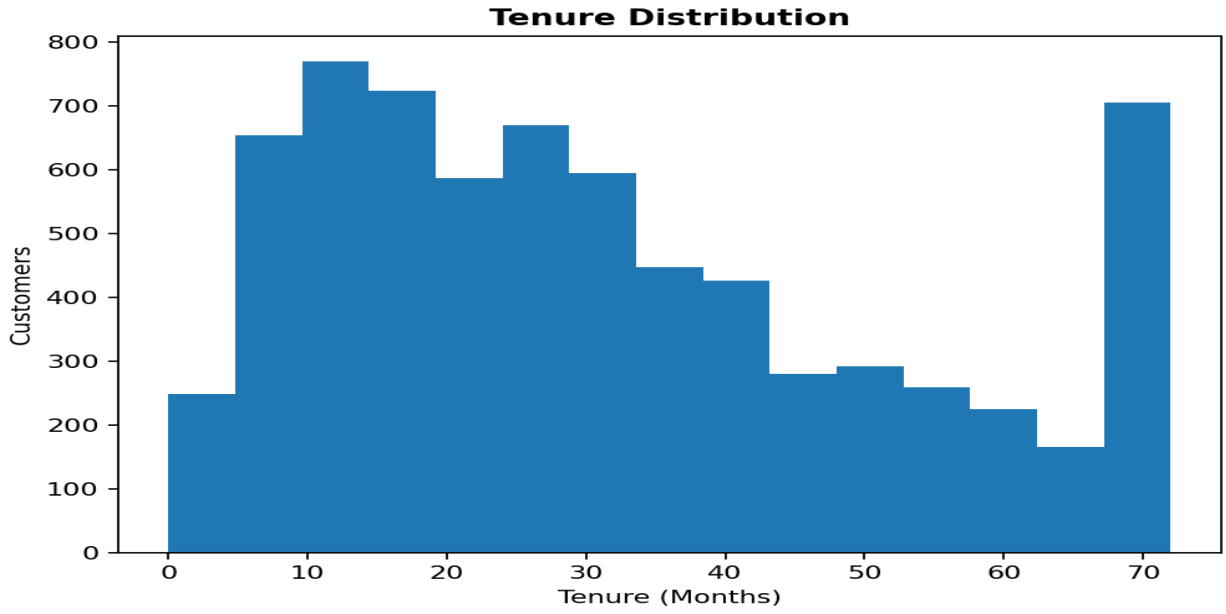

Customer Distribution by Contract
Customers
4000
3500
3000
2500
2000
1500
1000
500
0
Month-to-month
Two year
One year
Contract

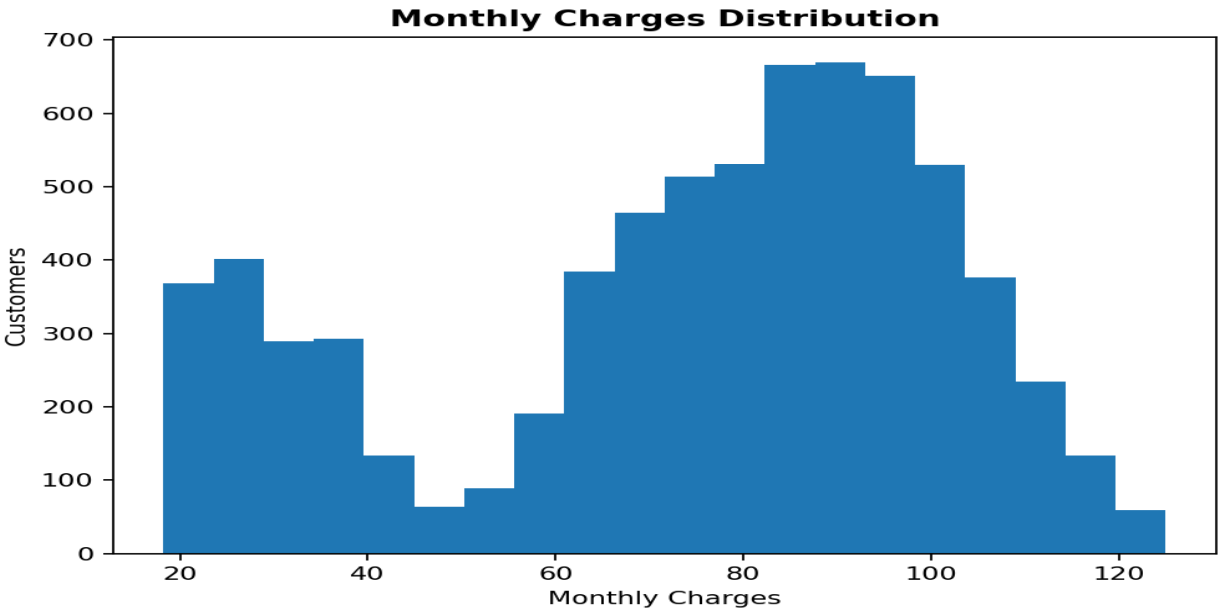

Churn Rate by Contract
Churn Rate (%)
40
30
20
10
0
Two year
One year
Month-to-month
Contract

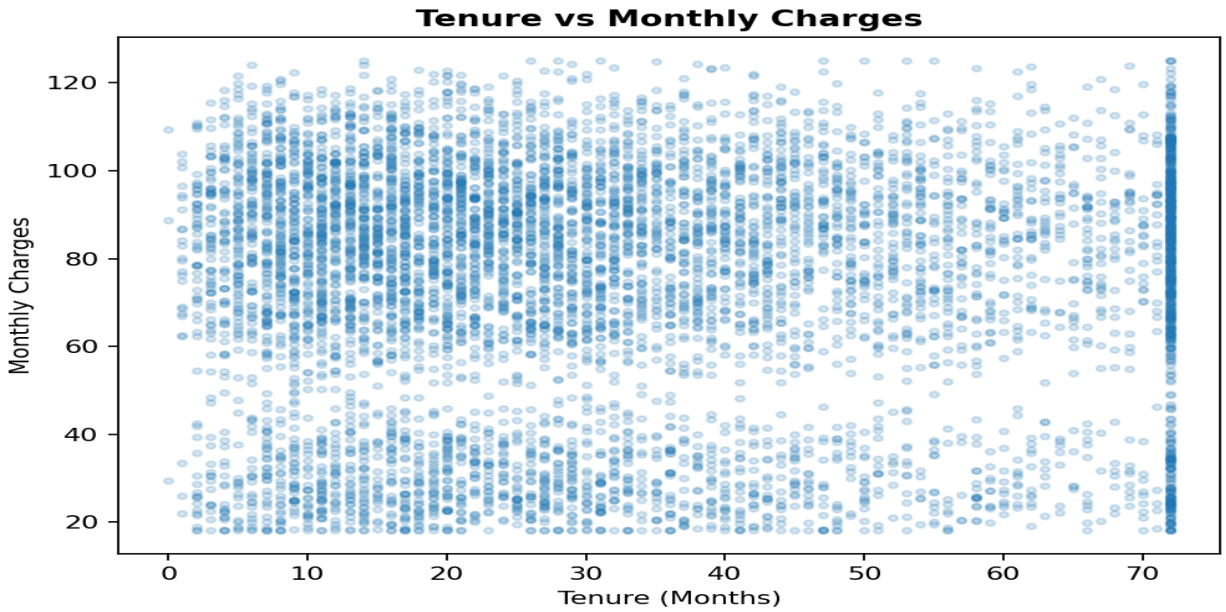

Churn Rate by Gender
Churn Rate (%)
30
25
20
15
10
5
0
Female
Male
Gender

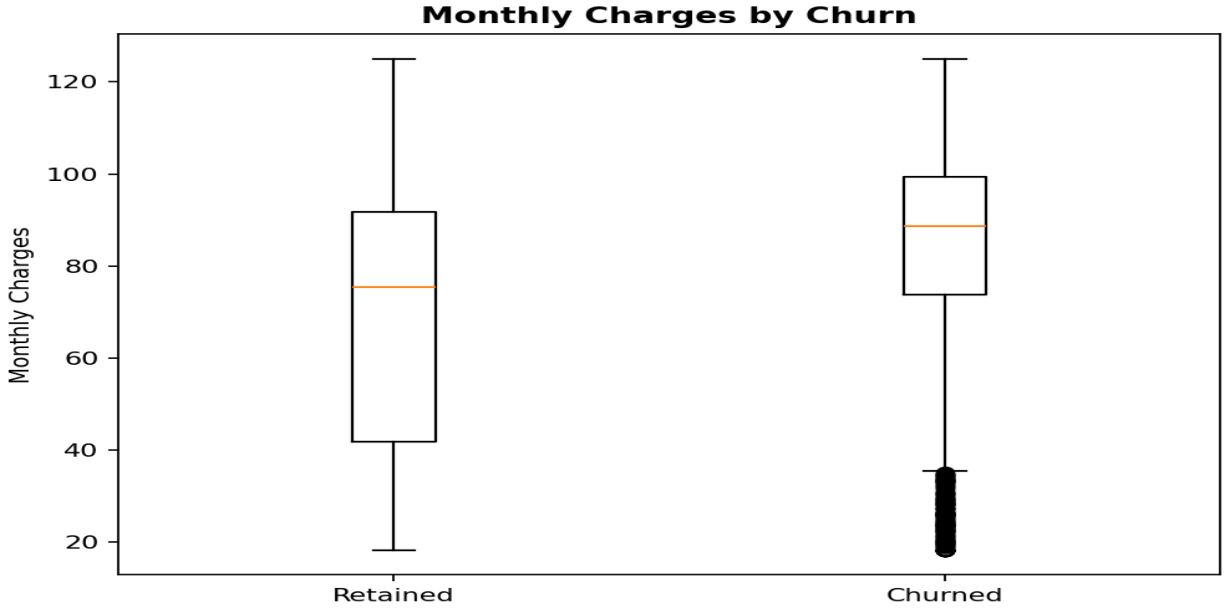

Churn Rate by Internet Service
Churn Rate (%)
0
10
20
30
40
No
DSL
Fiber optic
Internet Service

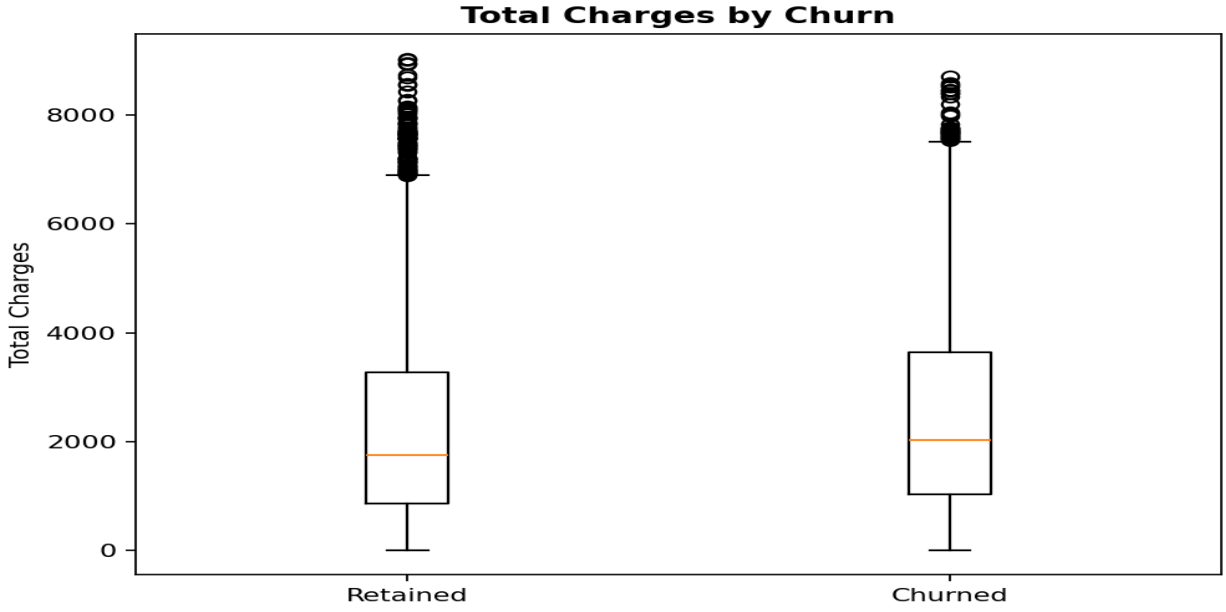

Churn Rate by Payment Method
Churn Rate (%)
0
5
10
15
20
25
30
35
Mailed check
Bank transfer (automatic)
Credit card (automatic)
Electronic check
Payment Method

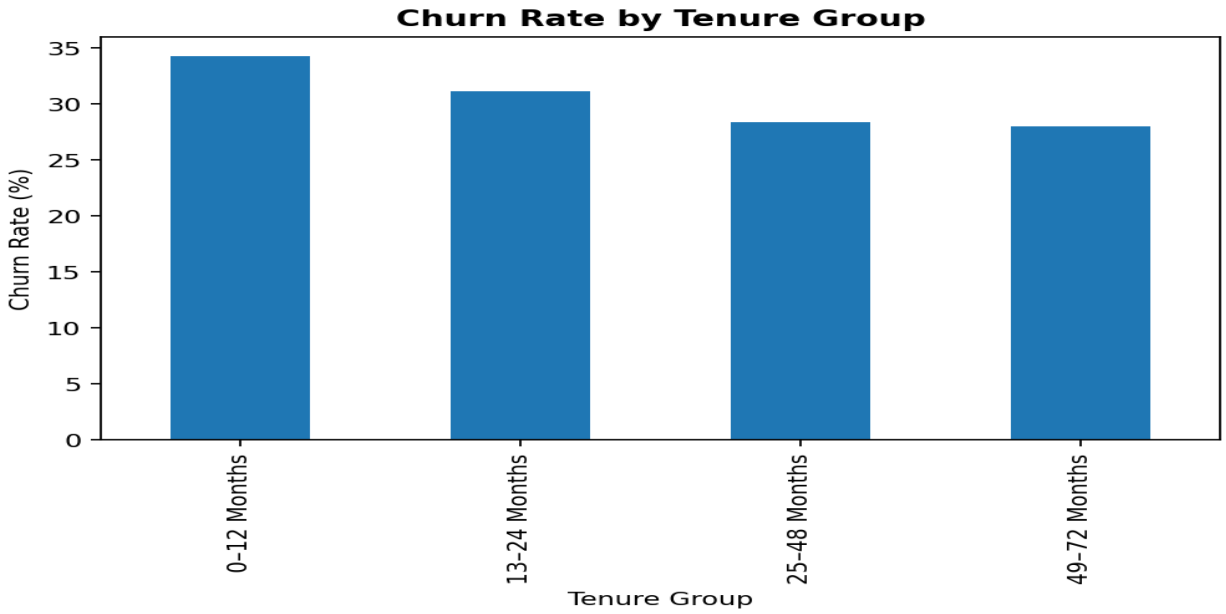

Tenure Distribution
Customers
0
100
200
300
400
500
600
700
800
0
10
20
30
40
50
60
70
Tenure (Months)

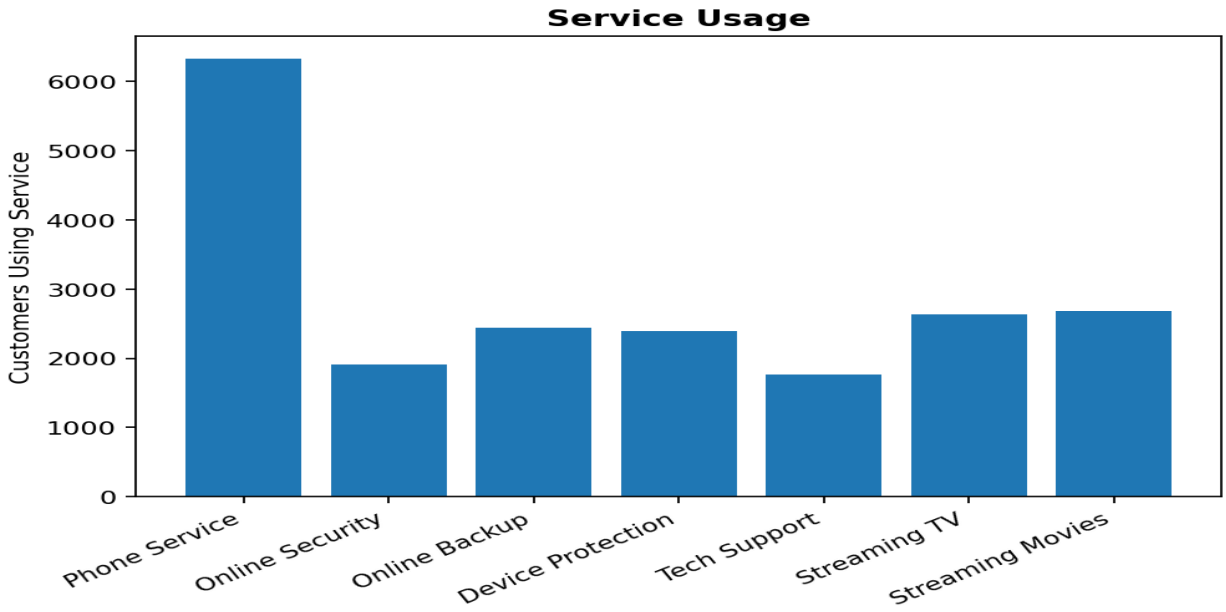

Monthly Charges Distribution
Customers
700
600
500
400
300
200
100
0
20
40
60
80
100
120
Monthly Charges

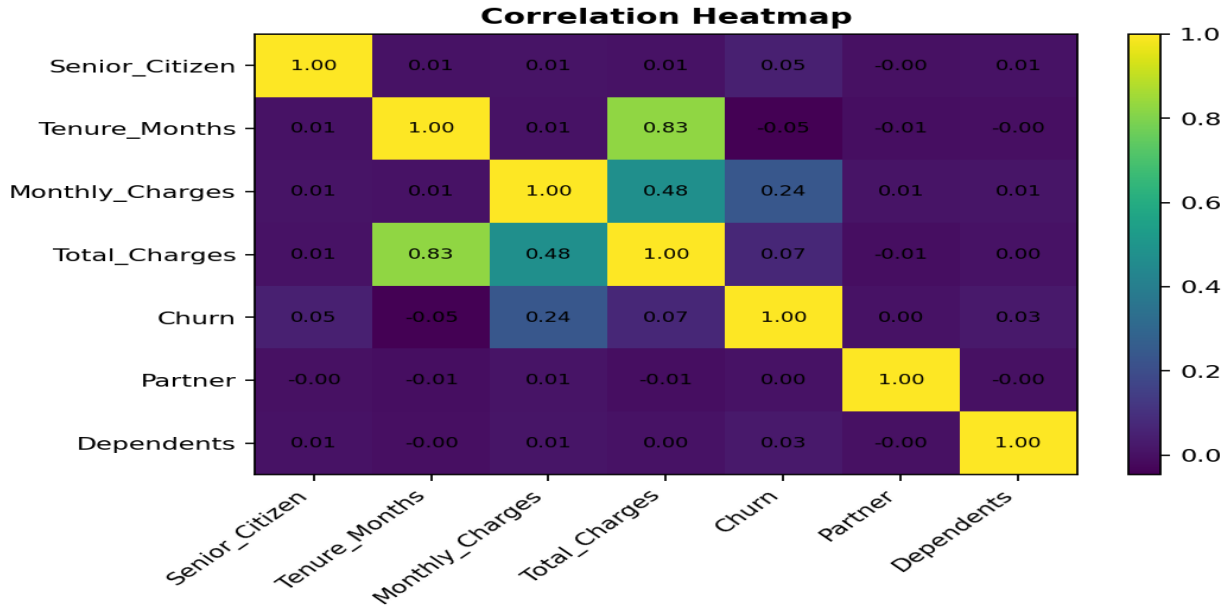

Tenure vs Monthly Charges
Monthly Charges
120
100
80
60
40
20
0
10
20
30
40
50
60
70
Tenure (Months)


Monthly Charges by Churn
Monthly Charges
120
100
80
60
40
20
Retained
Churned


Total Charges by Churn
Total Charges
8000
6000
4000
2000
0
Retained
Churned


Churn Rate by Tenure Group
Churn Rate (%)
0
5
10
15
20
25
30
35
0-12 Months
13-24 Months
25-48 Months
49-72 Months
Tenure Group





Power BI Requirements

Dashboard title: Customer Churn & Retention Analytics Dashboard. Required KPI cards: Total Customers, Churned Customers, Retained Customers, Churn Rate %, Average Monthly Charges and Average Tenure. Required visuals: Churn Distribution, Churn Rate by Contract, Churn Rate by Internet Service, Churn Rate by Payment Method, Customer Count by Tenure Group, Monthly Charges by Churn, Customer Distribution by Contract, Customer Distribution by Gender, Tenure vs Monthly Charges and Customer Details table/matrix. Required slicers: Gender, Contract, Internet Service, Payment Method, Senior Citizen and Churn.

DAX Measures

Total Customers; Churned Customers; Retained Customers; Churn Rate %; Average Monthly Charges; Average Tenure.

Business Insights

The current dataset has an overall churn rate of 30.12%. The highest-churn contract segment is Month-to-month (45.82%). The highest-churn payment method is Electronic check (37.73%). The highest-churn internet service is Fiber optic (43.77%). The tenure-group results and churned-versus-retained charge/tenure comparisons should be used to identify the most vulnerable customers and retention priorities.

Submission Checklist

Data cleaned; missing values handled; Python analysis completed; 14 Python visualizations completed; DAX measures documented; Power BI dashboard build guide prepared; business insights written; project report completed; cleaned dataset included; Python notebook included; README included. The native .pbix file still needs to be saved from Power BI Desktop because Power BI Desktop is not available in this environment.